SOLUTIONS - Exercise - Week 11: MATH1061/7861

(1) **(4 marks)** Determine which of the following is a subgroup of $(\mathbb{Z}_8, +)$. Justify your answer.
Recall we write 0 for the equivalence class [0] and similarly for other elements. In this notation, $\mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$.

(a) $\{0\}$.

This is a subgroup.
Observe that $0 + 0 = 0$, so it is closed, the identity is 0 and the inverse of 0 is 0.

(b) $\{0, 3\}$.

This is **not** a subgroup.
$3 + 3 = 6$, which is not in the set. Thus, it is not closed.

(c) $\{0, 2, 4, 6\}$.

This is a subgroup. There are two options: draw a Cayley table or prove the properties algebraically. A Cayley table is much easier.

+	0	2	4	6
0	0	2	4	6
2	2	4	6	0
4	4	6	0	2
6	6	0	2	4

From the Cayley table, it is clear that it is closed. The first row and column match the headers, so 0 is the identity. By looking at the locations of the 0s, we see that $0^{-1} = 0$, $2^{-1} = 6$, $4^{-1} = 4$ and $6^{-1} = 2$.

Alternatively, for all $x \in \{0, 2, 4, 6\}$ note that $x = 2k$ for some $k \in \mathbb{Z}$. Thus, for all $x, y \in \{0, 2, 4, 6\}$, take $x = 2k$ and $y = 2m$ for some $k, m \in \mathbb{Z}$. Then $x + y = 2k + 2m = 2(k + m)$. Considering this value modulo 8, we get that $2(k + m) = 8q + r$, and since $2(k + m)$ and $8q$ are both even, r must also be even. Thus, $2(k + m) \equiv r \pmod 8$ where $r \in \{0, 2, 4, 6\}$. Therefore, it is closed.
For all $x \in \{0, 2, 4, 6\}$, $0 + x = x = x + 0$. Thus, 0 is the identity. Observe that $0 + 0 = 0$, $2 + 6 \equiv 0 \pmod 8 \equiv 6 + 2$ and $4 + 4 \equiv 0 \pmod 8$. Therefore, $0^{-1} = 0$, $2^{-1} = 6$, $4^{-1} = 4$ and $6^{-1} = 2$.

MARKING:
1 mark for identifying that $\{0\}$ is a subgroup and justifying that it is closed, contains the identity, and contains all inverses
1 mark for identifying that $\{0, 3\}$ is not a subgroup (either by providing a counterexample to closure or inverse)
1 mark for noting that $\{0, 2, 4, 6\}$ is closed
1 mark for justifying why $\{0, 2, 4, 6\}$ contains inverses and the identity

(2) **(3 marks)** Solve the equation $10x + 9 = 6$ in the field $(\mathbb{Z}_{17}, +, \times)$. Show your working clearly.

Notice that $10 \times 12 \equiv 120 \equiv 1 \pmod{17}$. Using this we can calculate as follows:

$$\begin{aligned} 10x + 9 &\equiv 6 \pmod{17} \\ 10x &\equiv -3 \pmod{17} \\ 10x &\equiv 14 \pmod{17} \end{aligned}$$

From the hint: $10 \cdot 12 \equiv 1 \pmod{17}$, so we multiply by 12.

$$\begin{aligned} 12 \cdot 10x &\equiv 12 \cdot 14 \pmod{17} \\ 1 \cdot x &\equiv 168 \pmod{17} \\ x &\equiv 9 \cdot 17 + 15 \pmod{17} \\ &\equiv 15 \pmod{17} \end{aligned}$$

Therefore, $x = 15$ in the field.

MARKING:
1 mark for obtaining $10x \equiv 14 \pmod{17}$
2 marks for a correct reduced final answer (1 mark if the final answer is equivalent to 15 modulo 17, but is not reduced)

(3) **(3 marks)** How many arrangements of the letters in "MATHEMATICS" begin and end with the same letter?

There are 3 possible letters that can start the word: "A", "M" or "T". For any of these choices, there are 9 remaining letters to rearrange in the middle, with 2 remaining duplicate pairs of letters, so there are $9!/4$ ways to rearrange the middle 9 letters. Thus in total, there are

$$3 \times \frac{9!}{4} = 272160$$

arrangements.

MARKING:
1 mark for observing the 3 choices of initial and final letter pairs
1 mark for identifying that there are $9!/4$ ways to rearrange the middle 9 letters
1 mark for a correct final answer (either as a number, or as an equation)